

Clinical Trials Utilizing LLM-Based Generative AI

Hyon-Chel Jung*, Kun-Soo Shin**, Ho-Dong Kim***, Sung-Bin Park****

*Research Professor, Dept. of Institute of Artificial Intelligence and Big Data in Medicine, Yonsei University
Wonju College of Medicine, Korea

**Researcher, Dept. of Yonsei University Future Medical Industry Cooperation Group, Korea

***Executive Vice President, Head of AI, Corporate Research Institute, Solbit Co., Ltd., Korea

****Professor, Dept. of Precision Medicine, Yonsei University Wonju College of Medicine, Korea

[Abstract]

This study explores the improvement of work efficiency and expertise by applying Private LLM based on Large Language Model (LLM) to the field of clinical trials in medical devices. The Private LLM system provides sophisticated and accurate answers based on clinical data and shows its potential for use in various applications such as decision support, clinical expert activity assistance, new content generation, and problem solving. The study consists of the following four main steps. First, data specific to clinical trials of medical devices are collected, preprocessed, and organized into a learnable format. Second, based on open-source LLM models such as LaMA, PEFT (LoRA) and RAG techniques are applied to build a customized private LLM Q&A system for a specific clinical domain. Third, it realizes expert-level Q&A function by utilizing the established system and solves complex questions and problems that arise during clinical trial operation. Finally, by evaluating the performance of the system, we propose a direction to increase the efficiency and reliability of clinical trial operation and medical device development. As a result of the study, the Private LLM system has outperformed the existing methodology in supporting task automation and precise decision making. In particular, the ability to provide accurate answers to questions from domain experts and to generate new clinical criteria and insights shows the potential to become an innovative tool in clinical trial operation of medical devices. This confirmed the practical applicability of Private LLM in precision medical care, automation of clinical trial management, and a Q&A system based on domain knowledge.

▶ **Key words:** Large Language Models(LLMs), Generative AI, Clinical Trials, Medical Devices, Data Analysis

• First Author: Hyon-Chel Jung, Corresponding Author: Sung-Bin Park

*Hyon-Chel Jung (bravojhc@yonsei.ac.kr), Dept. of Institute of Artificial Intelligence and Big Data in Medicine, Yonsei University Wonju College of Medicine

**Kun-Soo Shin (tlsrjstn788@naver.com), Dept. of Yonsei University Future Medical Industry Cooperation Group

***Ho-Dong Kim (hodong.kim@solbit.kr), Corporate Research Institute, Solbit Co., Ltd.

****Sung-Bin Park (sung.b.park@gmail.com), Dept. of Precision Medicine, Yonsei University Wonju College of Medicine

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